Remember that the ‘players’ in the tournament were not humans but computer programs, preprogrammed strategies. Their human authors played the same role as genes programming bodies (think of Chapter 4’s computer chess and the Andromeda computer). You can think of the strategies as miniature ‘proxies’ for their authors. Indeed, one author could have submitted more than one strategy (although it would have been cheating—and Axelrod would presumably not have allowed it—for an author to ‘pack’ the competition with strategies, one of which received the benefits of sacrificial cooperation from the others).

Some ingenious strategies were submitted, though they were, of course, far less ingenious than their authors. The winning strategy, remarkably, was the simplest and superficially least ingenious of all. It was called Tit for Tat, and was submitted by Professor Anatol Rapoport, a well-known psychologist and games theorist from Toronto. Tit for Tat begins by cooperating on the first move and thereafter simply copies the previous move of the other player.

How might a game involving Tit for Tat proceed? As ever, what happens depends upon the other player. Suppose, first, that the other player is also Tit for Tat (remember that each strategy played against copies of itself as well as against the other 14). Both Tit for Tats begin by cooperating. In the next move, each player copies the other’s previous move, which was COOPERATE. Both continue to COOPERATE until the end of the game, and both end up with the full 100 per cent ‘benchmark’ score of 600 points.

Now suppose Tit for Tat plays against a strategy called Naive Prober. Naive Prober wasn’t actually entered in Axelrod’s competition, but it is instructive nevertheless. It is basically identical to Tit for Tat except that, once in a while, say on a random one in ten moves, it throws in a gratuitous defection and claims the high temptation score. Until Naive Prober tries one of its probing defections the players might as well be two Tit for Tats. A long and mutually profitable sequence of cooperation seems set to run its course, with a comfortable 100 per cent benchmark score for both players. But suddenly, without warning, say on the eighth move, Naive Prober defects. Tit for Tat, of course, has played COOPERATE on this move, and so is landed with the sucker’s pay-off of o points. Naive Prober appears to have done well, since it has obtained 5 points from that move. But in the next move Tit for Tat ‘retaliates’. It plays DEFECT, simply following its rule of imitating the opponent’s previous